MSG No. 100.177 P_C4bm [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4'm'm

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,4,7,8,10,11,13,14]
2	$\left[\frac{1}{2},\ 0,\ z\right]$	[2,3,5,6,9,12,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 4m'm'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[5,6,7,8,9,10,11,12]

Table 3: Wyckoff site: 4c, site symmetry: 2.mm

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[1,8,12,15]
2	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[2,5,11,14]
3	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[3,6,10,13]
4	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[4,7,9,16]

Table 4: Wyckoff site: 8d, site symmetry: .m'.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,13]
2	$[-y,\ 0,\ z]$	[2,15]
3	$[y,\ 0,\ z]$	[3,16]
4	$[0,\ -y,\ z]$	[4,14]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[5,9]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[6,12]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z\right]$	[7,10]
8	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z\right]$	[8,11]

Table 5: Wyckoff site: 8e, site symmetry: ..m

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,8]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,5]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,6]
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[4,7]
5	$\left[x+\frac{1}{2},\ x,\ z\right]$	[9,16]
6	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[10,13]
7	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[11,14]

8	$\left[\frac{1}{2}-x,\ -x,\ z\right]$	[12,15]

Table 6: Wyckoff site: 8f, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$[-x,\ x,\ z]$	[2,13]
3	$[x,\ -x,\ z]$	[3,14]
4	$[-x,\ -x,\ z]$	[4,15]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[5,10]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[6,11]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[7,12]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[8,9]

Table 7: Wyckoff site: 16g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[5]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[6]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[7]
8	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[10]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[12]
13	$[-x,\ y,\ z]$	[13]
14	$[x,\ -y,\ z]$	[14]
15	$[-y,\ -x,\ z]$	[15]
16	$[y,\ x,\ z]$	[16]